

Hong Vinh Vu

✉ yuvinh11@gmail.com 📍 Hanoi, VN 🌐 vu-vinh 🌐 vu-vinh

Education

Hanoi University of Science and Technology, Control and Automation Engineering Hanoi, Vietnam
 • Description 1. 2024 – 2028
 • Description 2.

Experience

Multi-Agent System Control (MASC) Research Laboratory, SEEE, HUST., Research Assistant HUST, VN
 2026 – 2028
 Applied Optimal Control Algorithms and Intelligent Control Algorithms to optimize trajectory planning for Quadrotors. 2 years
 • Developed a hybrid flight control framework by integrating PyTorch-based Deep Reinforcement Learning (DRL) for high-level dynamic trajectory planning with Model Predictive Control (MPC) and ANFIS for optimal trajectory tracking and nonlinear disturbance compensation.

Mechatronics Engineering Group (MEG), Motion Control and Applied Robotics Laboratory (MoCAR LAB), SEEE, HUST., Research Assistant HUST, VN
 Kinematic Modeling and Control of a 6WD-6WS Mobile Robot based on ROS2. Mar 2025 – 2026
 1 year
 • Developed mathematical kinematic/dynamic models and motion control algorithms for a 6WD-6WS (6-Wheel Drive, 6-Wheel Steering) omnidirectional vehicle, implementing custom ROS2 navigation nodes, Gazebo physics simulations, and embedded MCU firmware for real-time trajectory tracking and multi-mode steering coordination.

Awards

Nobel Prize in Physics Nov 1921
 The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'
 Royal Swedish Academy of Sciences
www.nobelprize.org/prizes/physics/1921/einstein/biographical

Publications

Zur Elektrodynamik bewegter Körper
 It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.
 Albert Einstein
en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Skills

Physics

Languages

German
 Native speaker

English
 Fluent

Interests

Physics

Certificates

Machine Learning	Jan 2018
Quantum Computing	Jan 2018
Quantum Information	Jan 2018

Projects

Quantum Computing Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers. • Quantum Teleportation • Quantum Cryptography	Jan 2018 – Jan 2018
--	---------------------

References

Professor John Doe

Professor Jane Smith